Tensors and Multilinear Algebra: An Approach To Kronecker Product and Applications in Image Representations

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1 Introduction

Applying basic linear algebra concepts, such as vectors and matrices, as a foundation for working with various datasets is essential. Nevertheless, some of us may question whether these concepts are sufficient to represent higher-dimensional data structures, such as colored images. Indeed, these standard linear algebra concepts are insufficient for analyzing and manipulating the complex structures of these higher-dimensional data.

This paper will discuss multilinear algebra and tensors, specifically Kronecker products, which are established on traditional linear algebra concepts such as linear transformations and dual spaces. In addition, this paper will discuss the process of rescaling an image and the applications of rank-n tensors in representing colored image datasets. To simplify the reading of this paper, vector spaces over $\mathbb{R}$ will be considered exclusively.

2 Introductory Linear Algebra Concepts

Before discussing multilinear algebra and tensors, recalling some of the essential standard linear algebra concepts is crucial. The foundation of multilinear algebra and tensors are built upon the fundamental concepts of linear algebra, such as linear transformations and dual spaces. The following sections will discuss these basic linear algebra concepts.

2.1 Linear Transformations

Linear transformations, or often called *linear maps*, are defined by the following.

Definition 1 (Linear Transformation (Axler & Ribet, 2015))**.** *A linear transformation from the vector space V to W is a map $T : V \to W$ with the following properties:*

1. *(addition) $T(u + v) = Tu + Tv$ for all $u, v \in V$,*
2. *(scalar multiplication) $T(\lambda v) = \lambda(Tv)$ for all $\lambda \in \mathbb{R}$ and $v \in V$.*

Note that for the set of all linear transformations from V to W, we use the notation $\mathcal{L}(V, W)$. In addition, we also have the operations of addition and scalar multiplication of the linear transformations $S, T \in \mathcal{L}(V, W)$ defined by

$$(S + T)(v) = Sv + Tv \quad \text{and} \quad (\lambda T)(v) = \lambda(Tv),$$

and $\mathcal{L}(V, W)$ is therefore a vector space.

Fact 1. *Suppose that V and W are finite-dimensional, then $\mathcal{L}(V, W)$ is finite-dimensional and*

$$dim\ \mathcal{L}(V, W) = (dim\ V)(dim\ W)$$

It is also important to talk about the linear transformations from a vector space V to itself, which is called an *operator*, and the notation $\mathcal{L}(V)$ denotes the set of all operators on V (Axler & Ribet, 2015). In other words, we have

$$\mathcal{L}(V) = \mathcal{L}(V, V).$$

2.2 Dual Space

The dual space of V is defined by the following

Definition 2. *The dual space of V, denoted V', is the vector space of all linear transformations from V to $\mathbb{R}$. In other words, we have $V' = \mathcal{L}(V, \mathbb{R})$*

In addition, the dual of a dual space is called a *bidual* of V, which in other words we have $(V')' = V$ ("Multilinear Algebra and Applications", 2019). Now, we want to introduce the *dual basis*, which is indeed a basis of V'. Given a basis $v_1, \ldots, v_n$ of V, then the dual basis of $v_1, \ldots, v_n$ is the list $\phi_1, \ldots, \phi_n$ of elements of V', where each ϕ_j is a linear transformation from V to $\mathbb{R}$ defined by the following:

$$\phi_j(v_k) = \delta_{kj} = \begin{cases} 1, & \text{if } k = j, \\ 0, & \text{if } k \neq j. \end{cases}$$

where δ_{ij} is the Kronecker delta.

Theorem 1. *Suppose V is finite-dimensional, then v' is also finite-dimensional and dim V' =dim V.*

Proof. By Definition 2, V' is the vector space of all linear transformations $\mathcal{L}(V, \mathbb{R})$. Therefore, we have

$$\begin{aligned} \text{dim } V' &= \text{dim } \mathcal{L}(V, \mathbb{R}) \\ &= (\text{dim } V)(\text{dim } \mathbb{R}) \end{aligned}$$

which follows from Fact 1. Since dim $\mathbb{R} = 1$, this implies that

$$\text{dim } \mathbb{V}' = \text{dim } V$$

as desired. □

The concept of dual space will later be used in the following sections in which we will introduce tensors.

3 Tensors

Tensors are geomteric object which holds linear transformation relations to another object in space and can be represented by multi-dimensional arrays. The following figures ("Multilinear Algebra and Applications", 2019) are the array representations of rank-n ($n \leq 3$) tensors:

$$\boxed{T_1 \quad \dots \quad T_n}$$

(a) Rank 1 tensor, denoted by T_i

$$\boxed{\begin{matrix} T_{11} & \dots & T_{1n} \\ \vdots & \dots & \vdots \\ T_{n1} & \dots & T_{nn} \end{matrix}}$$

(b) Rank 2 tensor, denoted by T_{ij}

Figure 1: Rank 1 and rank 2 tensors

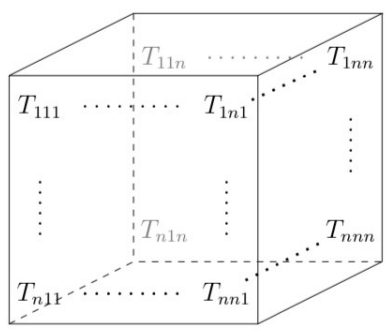

Figure 2: Rank 3 tensor, denoted by T_{ijk} ("Multilinear Algebra and Applications", 2019)

3.1 Tensors as Multilinear Transformations

The concept of a multilinear transformation is similar to the concept of a linear transformation, except this time we consider the collection of vector spaces $V_1, \dots, V_n$.

Definition 3. *A n-linear transformation is defined by a map $T^* : V_1 \times \dots \times V_n \to \mathbb{R}$ with the following properties:*

1. $T^*(v_1, \dots, v_i + v_j, \dots, v_n) = T^*(v_1, \dots, v_i, \dots, v_n) + T^*(v_1, \dots, v_j, \dots, v_n)$ *for $v_k \in V_k$ and $k \in [1, n]$,*

2. $T^*(v_1, \ldots, \lambda v_i, \ldots, v_n) = \lambda(T^*(v_1, \ldots, v_i, \ldots, v_n))$ *for* $v_k \in V_k$ *and* $k \in [1, n]$.

Tensors are the more specific type of multilinear transformation. Here, we define a tensor of order (p, q) on vector space V as a rank $p + q$ function

$$\tilde{T} : \underbrace{V' \times \cdots \times V'}_{p \text{ times}} \times \underbrace{V \times \cdots \times V}_{q \text{ times}} \to \mathbb{R}$$

for all positive integers p and q (Shurbert, 2014).

It is also important to talk about the set of all tensors of order (p, q), which is defined as follows:

Definition 4 (("Multilinear Algebra and Applications", 2019))**.** *The set of all tensors of order* (p, q) *on a vector space* V *is denoted by*

$$\mathcal{T}^p_q(V) := \{\textit{all order } (p, q) \textit{ tensors on } V\}$$

Using the fact that we have the bidual $(V')'$ of v and Definition 2, we can conclude that

$$\mathcal{T}^0_1(V) = \mathcal{L}(V, \mathbb{R}) = V' \quad \text{and} \quad \mathcal{T}^1_0(V) = \mathcal{L}(V', \mathbb{R}) = (V')' = V$$

where $\mathcal{T}^0_1(V)$ is a *covariant* tensor set and $\mathcal{T}^1_0(V)$ is a *contravariant* tensor set.

3.2 Kronecker Product

The Kronecker product, also known as the tensor product, is an operation that combines two matrices to produce a larger matrix. The Kronecker product is denoted by the symbol $\otimes$ and is defined as follows:

Definition 5. *Given two matrices* A *and* B *of sizes* $m \times n$ *and* $p \times q$*, respectively, the Kronecker product* $A \otimes B$ *is the* $mp \times nq$ *block matrix defined as*

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{pmatrix}$$

where a_{ij} *is the element of* A *in the* i*-th row and* j*-th column, and* $a_{ij}B$ *denotes the* $p \times q$ *matrix obtained by multiplying* a_{ij} *by every element of* B*.*

In other words, the Kronecker product constructs a matrix whose (i, j)-th block is obtained by multiplying the (i, j)-th entry of A by the entire matrix B. This operation can be thought of as a way of creating a block diagonal matrix, where the blocks are scaled copies of the matrix B, and the scaling factors are given by the entries of A.

Example 1 ((Whitcomb, 2020)). *Given*

$$A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}_{2\times 2} \quad B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}_{2\times 3}$$

the Kronecker product $A \otimes B$ is

$$\begin{aligned} A \otimes B &= \begin{pmatrix} 1\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} & 2\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \\ 0\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} & -1\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2 & 3 & 2 & 4 & 6 \\ 4 & 5 & 6 & 8 & 10 & 12 \\ 0 & 0 & 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & -4 & -5 & -6 \end{pmatrix}_{4\times 6} \end{aligned}$$

4 Resizing RGB Images Using Kronecker Product

To scale the size of an RGB $n \times m$ image by a factor of r using the Kronecker product, we can create an $r \times r$ matrix of ones, denoted by M, and then take the Kronecker product of this matrix with the original image.

Let the original image be denoted by the tensor X of size $m \times n \times k$ as follows:

$$X = \begin{pmatrix} (a_{111}\ a_{112}\ a_{113}) & (a_{121}\ a_{122}\ a_{123}) & \dots & (a_{1m1}\ a_{1m2}\ a_{1m3}) \\ (a_{211}\ a_{212}\ a_{213}) & (a_{221}\ a_{222}\ a_{223}) & \dots & (a_{2m1}\ a_{2m2}\ a_{2m3}) \\ \vdots & \vdots & \ddots & \vdots \\ (a_{n11}\ a_{n12}\ a_{n13}) & (a_{n21}\ a_{n22}\ a_{n23}) & \dots & (a_{nm1}\ a_{nm2}\ a_{nm3}) \end{pmatrix}_{m\times n\times k}$$

where the last dimension is a dimension 3 array that represents the RGB channels. Next, we create an $r \times r$ matrix of ones, denoted by M, as follows:

$$M = \begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 1 \end{pmatrix}_{r\times r}$$

and we take the Kronecker product $X \otimes M$ to obtain:

$$
\begin{aligned}
X \otimes M &= \begin{pmatrix} (a_{111}\ a_{112}\ a_{113}) \begin{pmatrix} 1 & 1 & \dots \\ 1 & 1 & \dots \\ \vdots & \ddots & \vdots \end{pmatrix}_{r \times r} & \dots \\ & \vdots & \vdots \end{pmatrix} \\
&= \begin{pmatrix} (a_{111}\ a_{112}\ a_{113}) & (a_{111}\ a_{112}\ a_{113}) & \dots & (a_{121}\ a_{122}\ a_{123}) & \dots \\ (a_{111}\ a_{112}\ a_{113}) & (a_{111}\ a_{112}\ a_{113}) & \dots & (a_{121}\ a_{122}\ a_{123}) & \dots \\ \vdots & \vdots & \ddots & \vdots & \vdots \end{pmatrix}
\end{aligned}
$$

where $X \otimes M$ is the rescaled image M by the scalar r and have the size $rm \times rn \times rk$. Therefore, we have the rescaled image $X \otimes M$ by a scalar r, as desired.

5 Applications: Rank-n Tensors in Image Dataset Representations

A tensor of rank 0 is a scalar, which is a singular value with no direction or orientation. A rank-0 tensor can represent the value of a single pixel in the context of images. As a grayscale image is basically a two-dimensional array of pixel values, each pixel can be represented as a rank-zero tensor. A tensor of rank 1 is a vector, which is an array of values with direction and magnitude. A rank-1 tensor can represent a row or column of pixels in an image in the context of images. Each vector element represents the value of a single pixel. A rank-2 tensor is a matrix, which is a collection of values with direction and orientation. A rank-2 tensor can represent the entire 2D array of intensity values that form an image in the context of images. Each matrix element represents the value of a single pixel. In operations such as resizing or cropping an image, where the entire image must be altered, rank-2 tensors are widely employed.

A rank-3 tensor is a three-dimensional array that consists of layered matrices. A rank-3 tensor can represent a collection of images in the context of images, where each matrix corresponds to a single image, and it can also represent a colored (RGB) image. A rank-4 tensor is a 4D array, which is a collection of 3D arrays layered on top of one another. A rank-4 tensor can represent a collection of 3D arrays in the context of images, where each 3D array corresponds to a collection of images. In order to increase the effectiveness of training a machine learning model, rank-4 tensors are frequently employed in operations such as batch processing, in which a set of images are processed simultaneously.

6 Conclusion

In image representation and analysis, tensors and the Kronecker product play a crucial role. Tensors allow images to be depicted as multidimensional arrays, which makes them suitable for efficient processing and analysis. The Kronecker product is a valuable instrument for manipulating tensors and generating larger tensors from smaller ones, which can be applied to a variety of image processing tasks. The significance of tensors and the Kronecker product in the representation and analysis of images extends beyond computer vision. There are a variety of applications for them in disciplines such as data science and machine learning.

The ability to represent and manipulate complex data structures, such as images, with tensors and the Kronecker product is crucial for the growth of research in these disciplines. The significance of tensors and the Kronecker product in the representation and analysis of images cannot be highlighted. Their applications are limited only by the possibilities that exist, as they are highly effective instruments that facilitate efficient image processing, analysis, and manipulation.

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